

Digital Forensics Report: File Recovery and Metadata Analysis

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1. Forensic Investigation Overview

Case Summary

The primary objective of this investigation was to analyze a forensic disk image (`fat32_forensic.dd`) to identify deleted files, detect timestamp anomalies indicating potential tampering, and perform both manual and automated file recovery (carving).

Feature	Details
	FAT32 (File Allocation Table 32)

Feature	Details
File System Type	
Source Image	fat32_forensic.dd
Core Objectives	File Recovery, Metadata Analysis, Tamper Detection

Forensic Insight: File recovery is critical because deleted data often contains the "smoking gun" evidence in an investigation. Even when a user deletes a file, the data typically remains in the unallocated space until overwritten, providing a window for forensic reconstruction.

2. File System Metadata Analysis

A detailed analysis of the file system structure was conducted to establish the parameters for data recovery.

Key Metadata Findings

Attribute	Value
Sector Size	512 bytes
Cluster Size	512 bytes (1 sector per cluster)

Attribute	Value
Total Sector Range	0 - 102399 sectors
Data Area Start	Sector 1608

Criticality of Metadata: Metadata provides the "map" of the drive. Without knowing the cluster size and sector layout, an analyst cannot calculate the exact byte offsets needed to carve files out of the raw binary data.

3. File Recovery Process

Locating Deleted Files

Using forensic tools such as `fls`, `fsstat`, and `istat`, several deleted files were identified within the image.

File Name	Entry Number	Size (Bytes)	Starting Sector
IMG_2019.jpg	4	3,190,011	32990
IMG_0117.png	6	Not Specified	Not Specified

Analyzing Suspicious Timestamps

During the analysis of `IMG_2019.jpg`, a significant anomaly was detected:

- **Anomalous Written Timestamp:** 2036-01-01

- **Assessment:** This indicates "**Time-Stomping**" or system clock manipulation. Since the date is in the future, it is a clear indicator that the file's temporal metadata has been altered to obfuscate the true timeline of events.
- **Impact:** Suspicious timestamps cast doubt on the integrity of the file and suggest a sophisticated user attempted to hide activity.

4. Byte Offset Calculation for File Carving

To ensure a bit-for-bit accurate recovery of `IMG_2019.jpg`, the following calculations were performed:

Step-by-Step Calculation

1. Start of File (SOF):

Decimal: $32990 \text{ (Sector)} \times 512 \text{ (Bytes)} = 16,890,880$

Hexadecimal: `0x101bc00`

2. End of File (EOF):

Decimal: $16,890,880 \text{ (SOF)} + 3,190,011 \text{ (Size)} = 20,080,891$

Hexadecimal: `0x13268fb`

Technical Comparison: In FAT32, the metadata is simpler but more prone to fragmentation. In NTFS, we would look at the Master File Table (\$MFT) for "Resident" or "Non-resident" attributes, which provides more granular detail but adds complexity to the calculation.

5. File Recovery and Integrity Validation

Recovery Methodology

- **CLI Recovery (icat):** Executed `sudo icat fat32_forensic.dd 4 > recovered_jpg.jpg`. The file was successfully extracted and verified as a valid JPEG.
- **Python Carving:** Used `file_carver.py` with an `offsets.txt` file. Initial conversion errors (header related) were resolved by refining the script's input handling.

MD5 Hash Verification

Integrity was verified by comparing the MD5 hash of the recovered file against the original provided hash.

File	MD5 Hash Value
Recovered File	b3ab520fa902f0f21d3e77d399aabde2
Original Reference	b3ab520fa902f0f21d3e77d399aabde2

File	MD5 Hash Value
Result	MATCH - Integrity Verified

6. Reflection and Forensic Insights

Challenges and Solutions

- **Challenge:** Python script failed with `invalid literal for int()`.
- **Resolution:** Identified that the script attempted to convert the string header "StartOffset" into an integer. Re-ran the script focusing on raw numeric data, ensuring logic parity with manual `icat` results.

FAT32 vs. NTFS Comparison

Feature	FAT32	NTFS
Metadata Source	Directory Entry Table	Master File Table (\$MFT)
Time Resolution	2 seconds	100 nanoseconds
Carving Complexity	Lower (Linear)	Higher (Attributes/Fragmentation)

Final Forensic Conclusions

1. **Metadata Importance:** It is the foundation of evidence; it tells us how a file was handled, not just what is in it.
2. **Timestamps:** They are the primary tool for reconstructing the sequence of events (the "Timeline").
3. **Automation:** Automated carving with Python is significantly more efficient than manual recovery, reducing human error in complex conversions.